

VanniKawachh: A Distributed AI Acoustic Intelligence and Autonomous Drone Response Network for Women Safety

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Abstract—Crimes against women concentrate in surveillance and cellular dead zones—dark streets, forest stretches, campus outskirts—and the prevailing safety technology burdens the victim: panic apps, wearables, and carried devices protect only the moments in which a charged, reachable device is in the victim’s hand. We present VanniKawachh, a three-tier network that moves the trigger burden from the victim to the infrastructure: her voice is the trigger. Solar-powered, pole-mounted sensing nodes (ESP32-S3 + INMP441) screen every audio frame on-device with a lightweight MFCC + CNN model (Stage 1, <50 ms budget, recall-tuned), so no continuous audio ever leaves a pole. A Raspberry Pi 5 hub confirms candidate events with PANNs deep audio tagging fused with PIR motion, ambient-light, and time-of-day evidence (Stage 2, precision-tuned), gating dispatch behind explicit verification and severity thresholds. Confirmed alerts—25-byte packets sealed with AES-128-CTR, truncated HMAC-SHA256, and a per-node replay counter, carrying surveyed node coordinates from a registry rather than a live GPS fix—travel over LoRa with no cellular dependency to a police dashboard, and simultaneously auto-dispatch an autonomous quadcopter that records evidence during hover, descends to 3 m, drops a first-aid kit (fail → return-to-launch), and returns. The response layer is our previously SITL-verified flight stack—verified mode transition, a landing-interlocked mission queue, and a debounced severity-ordered failsafe arbiter—carried over unchanged. The complete chain runs end-to-end with zero hardware: the Phase-0 rehearsal passed with fused severity 0.88 and kit release commanded at 3.1 m inside a 328 s SITL mission; the acceptance harness passes 8/8 checks with 0.4–0.6 m terminal accuracy; 68 automated unit cases pin the safety logic, packet cryptography, and dispatch gating. Stage-1 model training and hardware range measurements are Phase 1–2 work in progress; the development chain used clearly labelled heuristic stand-ins, and this paper reports no acoustic accuracy it has not measured.

Index Terms—women safety; acoustic event detection; TinyML; PANNs; sensor fusion; LoRa; AES-128; autonomous UAV; verified dispatch; SITL.

I. Introduction

Where does safety technology fail? Precisely where assaults happen: in locations that are camera-poor, patrol-poor, and often cellular dead zones, at moments when the victim cannot operate a device—hands occupied, phone snatched, discharged, or out of reach. The deployed answer to women’s safety has been to instrument the victim: panic

buttons, mobile apps, smart wearables. The best of these report high trigger accuracy, but their failure mode is structural rather than statistical—a device-borne trigger protects only the moments in which the device is present, charged, and reachable.

VanniKawachh (“voice-shield”) inverts the burden. Pole-mounted infrastructure listens; a scream, a cry, or a shouted “help” / “bachao”—signals the victim produces with no device at all—is the trigger. The engineering problem is then threefold: (i) detection must run on a solar pole budget without streaming audio anywhere; (ii) false alarms must be suppressed well enough that acting on an alert is defensible, because the action is a drone launch; and (iii) the alert path must work exactly where cellular does not, and must be unforgeable, because a spoofed alert weaponizes the response.

Our contributions are:

- 1) An end-to-end architecture—infrastructure sensing → two-stage verification → offline encrypted alerting → autonomous field response—where surveyed prior work covers at most one link (Section II).
- 2) A two-stage acoustic pipeline splitting recall and precision between an on-node TinyML screen and a hub-side PANNs + sensor-fusion confirmation, with tested dispatch gating (Section IV).
- 3) A secure LoRa alert protocol: sealed 25-byte packets resolved against a surveyed node registry, so position is never transmitted or trusted from the field (Section V).
- 4) A safety-verified autonomous response layer—our SITL-verified dispatch stack, extended with an evidence camera window and a rule-bounded first-aid kit drop (Section VI).
- 5) A zero-hardware full-chain validation (Phase 0) demonstrating the integrated system in software-in-the-loop simulation, with an explicit account of which numbers are measured and which await the hardware phases (Section VII).

II. Related Work

Our twelve-paper survey (2023–2026; entries [S1]–[S12], full bibliography in the group’s seminar record and the journal version) [13] partitions the field on one question: who carries the trigger?

Victim-carried devices (survey entries [S1], [S5], [S6], [S9], [S10], [S11] of [13])—apps, wearables, IoT panic buttons—report trigger-classification accuracies up to 97.5%, typically with GPS + GSM alerting. All fail identically when the device is absent, damaged, discharged, or unreachable, and inherit cellular dependence in exactly the locations that lack it.

Detection-only audio systems (entries [S2], [S3], [S4], [S7], [S8] of [13]) reach 92–95.5% scream detection with CNN–Transformer hybrids and transfer-learned backbones (InceptionV3, MobileNetV2) on mel-spectrograms—but on server-scale compute, evaluated on curated data, and stopping at classification: no location delivery, no dead-zone alerting, no response.

The gap is the chain itself: no surveyed system provides infrastructure sensing → verified offline alerting → autonomous response end to end. Each link is individually supported by adjacent literatures—TensorFlow Lite Micro keyword spotting [7] for the node, PANNs large-scale pretrained audio tagging [6] for the hub, LPWAN practice [8] for the alert path, and the open-source autopilot ecosystem [1], [2] for the response—and our prior flight-stack work supplies the response layer’s safety machinery, including the empirical finding that motivates the whole design philosophy: flight autopilots can silently reject mode commands while client libraries report success. VanniKawachh generalizes the resulting discipline—a claim is not a confirmation—to every layer: sound, packet, dispatch, and flight mode.

III. System Architecture

Fig. 1 traces one incident through the three tiers: the solar-powered sensing node (ESP32-S3 + INMP441, PIR + LDR context, Stage-1 hit → sealed alert over LoRa + 4 s clip over ESP-NOW/WiFi), the Raspberry Pi 5 hub (unseal, registry lookup, Stage-2 PANNs + fusion, police dashboard, POST /trigger), and the SITL-verified response drone (trigger API → queue → 13-state FSM with hover-record, 3 m kit drop, and RTL).

Three design decisions carry the architecture. Fixed nodes carry no live GPS: each pole is surveyed once at installation; the hub’s registry maps node_id → (lat, lon), so the radio carries two bytes of identity, and a node has no GPS to spoof, jam, or drain. LoRa carries the alert, never the audio: LoRa’s ~1–5.5 kbps effective throughput cannot move a clip, so the sealed alert goes over LoRa instantly while the 4 s verification clip follows over ESP-NOW/WiFi (~250 kbps, hundreds of metres LOS); if the clip never arrives within 8 s, the hub degrades to the Stage-1 confidence at a ×0.6 haircut—always logging, dispatching only on otherwise-strong evidence. The flight

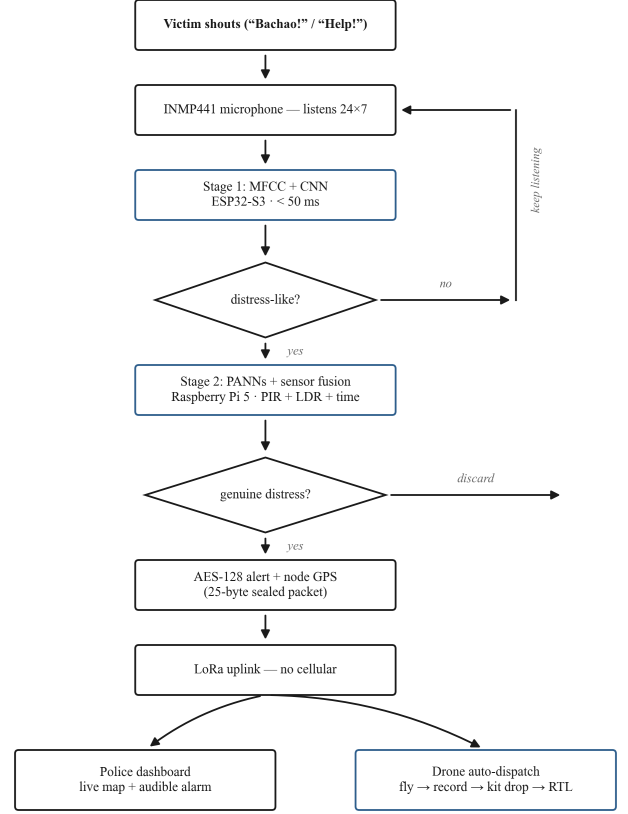


Fig. 1. The VanniKawachh chain, from acoustic event through Stage-1 on-node screening, Stage-2 hub verification and fusion, and sealed LoRa alerting to the parallel dashboard and drone-dispatch paths. Each tier verifies before it acts.

core is untouched: all v1 safety machinery carries over unchanged, which is why the response half of the system already works.

IV. The Two-Stage Acoustic Pipeline

Stage 1 (node, recall-tuned). The ESP32-S3 frames 16 kHz mono audio from the INMP441, extracts MFCCs, and runs a tiny quantized CNN (TensorFlow Lite Micro, micro_speech-class [7]) against the distress vocabulary—scream, cry, “help”/“bachao” keywords—within a <50 ms per-frame budget. The front end is the standard speech chain: each frame is pre-emphasised,

$$y[n] = x[n] - 0.97 x[n-1], \quad (1)$$

Hamming-windowed,

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad (2)$$

and its magnitude spectrum pooled by M triangular filters spaced uniformly on the mel scale

$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right), \quad (3)$$

giving log filterbank energies

$$e_j = \log \sum_k |X(k)|^2 H_j(k), \quad (4)$$

which a DCT-II decorrelates into the 13 cepstral coefficients used as the per-frame feature vector,

$$c_i = \sum_{j=1}^M e_j \cos \left[i \left(j - \frac{1}{2} \right) \frac{\pi}{M} \right], \quad i = 1, \dots, 13. \quad (5)$$

The CNN classifies each MFCC frame through a softmax output,

$$p_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}, \quad (6)$$

is trained with the cross-entropy loss

$$L = - \sum_i y_i \log p_i, \quad (7)$$

and deploys after int8 post-training quantization,

$$x_q = \text{round} \left(\frac{x}{s} \right) + z, \quad (8)$$

with per-tensor scale s and zero-point z —the step that fits the model into TFLM on the ESP32-S3. Frames that do not trip Stage 1 are discarded on the spot: nothing stored, nothing transmitted. Stage 1 exists to not miss; its false positives are expected and cheap because Stage 2 filters them. (The model itself is a hook in the current firmware; training and flashing it is Phase-1 work—see Section VII-C.)

Stage 2 (hub, precision-tuned). The hub re-scores the clip with PANNs [6]—the pretrained AudioSet tagging network (CNN14, or a lighter checkpoint on a slow Pi)—taking the summed probability over the distress-relevant AudioSet classes (screaming, shouting, yelling, crying, wailing, ...) as a distress score in $[0, 1]$. No bespoke training is required; the hub leans on AudioSet scale, which a locality Pi 5 can afford and a solar pole cannot. A labelled energy-heuristic fallback backend (loud + high-spectral-centroid + bursty) exists so the whole chain runs on any development machine; it is not a claim of accuracy and every result produced with it is marked as fallback.

Fusion. A night-time scream in a dark spot with motion nearby is a different animal from a daytime shout on a busy road. With a the Stage-2 audio score, c the Stage-1 confidence, $p \in \{0, 1\}$ the PIR motion flag, $d = 1 - L/255$ the darkness derived from the LDR level $L \in [0, 255]$, and $n \in \{0, 1\}$ the night indicator (1 during 20:00–06:00), the fused severity is

$$S = 0.60a + 0.15c + 0.10p + 0.08d + 0.07n, \quad (9)$$

with priority high at $S \geq 0.75$ or verified audio ($a \geq 0.6$) coinciding with PIR motion ($p = 1$). Dispatch requires both gates to pass,

$$\text{dispatch} \iff (a \geq \tau_v) \wedge (S \geq \tau_d), \quad \tau_v = 0.50, \tau_d = 0.60; \quad (10)$$

Scaled alert packet — 25 bytes total (one LoRa frame)

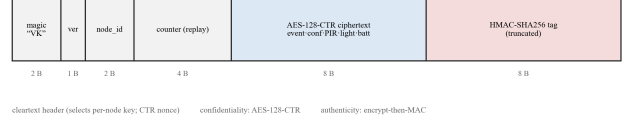


Fig. 2. The 25-byte sealed alert packet: cleartext header, AES-128-CTR ciphertext, and truncated HMAC-SHA256 tag—one LoRa frame.

below either threshold the incident is logged with a human-readable reasons trace and no drone flies. When the verification clip never arrives (Section III), the degraded audio score

$$a = 0.6c \quad (11)$$

is substituted before applying (9)–(10). The gating is pinned by automated tests, and the weights in (9) are prototype values to be tuned against Phase-1 bench data.

V. Secure LoRa Alerting

A spoofed packet would launch a drone; a replayed one would launch it at an attacker’s chosen time. Every alert is therefore a sealed 25-byte packet—one comfortable LoRa frame, laid out as shown in Fig. 2: a cleartext header (magic, version, node_id uint16, counter uint32), an 8-byte AES-128-CTR-encrypted payload (event class, Stage-1 confidence, PIR flag, LDR level, node battery), and an 8-byte MAC (HMAC-SHA256 over header + ciphertext, truncated).

Per-node keys derive from the master key K_m by keyed hashing (FIPS-197 AES, RFC 2104 HMAC [10]),

$$K_n = \text{HMAC-SHA256}(K_m, \text{"node:"} \parallel n)[0:16], \quad (12)$$

so provisioning needs only the master key and an id, and a captured node compromises one pole, revocable in the registry. The payload P is encrypted in CTR mode,

$$C = P \oplus E_{K_n}(\text{IV}), \quad (13)$$

with the counter block IV built from the cleartext header (magic $\parallel$ ver $\parallel$ node_id $\parallel$ counter)—unique per packet while the counter is monotonic—and authenticated encrypt-then-MAC,

$$\tau = \text{HMAC-SHA256}(K_n, \text{header} \parallel C)[0:8]. \quad (14)$$

The hub accepts a packet only when

$$\tau \text{ valid} \wedge \text{counter} > \text{counter}_{\text{last}}, \quad (15)$$

so bad MACs, unknown node ids, and replays are all rejected; forging the 64-bit truncated tag succeeds with probability 2^{-64} per attempt. Above the radio, the hub reaches the drone stack through the same token-authenticated, geofence-validated API as any operator—no privileged backdoor. Residual risk is stated: jamming is

a denial, not a spoof; multi-node corroboration and node-liveness monitoring are future answers.

The LoRa/WiFi split of Section III is quantitative physics, not preference. LoRa’s range comes from spreading—each symbol lasts

$$T_s = \frac{2^{\text{SF}}}{\text{BW}}, \quad (16)$$

and the receiver sensitivity floor

$$S_{\text{dBm}} = -174 + 10 \log_{10}(\text{BW}) + \text{NF} + \text{SNR}_{\text{min}} \quad (17)$$

drops below -120 dBm at high SF over $\text{BW} = 125$ kHz, buying kilometres from milliwatts. The same trade caps throughput: the standard LoRa time-on-air expression (Semtech SX127x datasheet [8]) gives ≈ 0.21 s for the 25-byte sealed alert at SF9, BW 125 kHz, CR 4/5, explicit header + CRC, whereas the 4 s clip (≈ 128 kB at ≈ 5.5 kb/s effective) would occupy the channel for minutes—hence alert over LoRa, clip over ESP-NOW/WiFi.

Privacy is by construction, not policy: continuous audio cannot leave a node because the transport cannot carry it and the clip path is event-gated in firmware; only event-triggered clips ≤ 5 s are ever transmitted, and the alert itself is encrypted.

VI. Safety-Verified Drone Response

The response layer is our SITL-verified dispatch stack, whose three mechanisms transfer intact. Verified mode transition: every flight-mode command—nominal or emergency—routes through a single routine that re-issues the request through layered MAVLink encodings (COMMAND_LONG/MAV_CMD_DO_SET_MODE plus legacy SET_MODE) on a 700 ms cadence until the autopilot’s own HEARTBEAT-derived mode confirms adoption, with a cross-action fallback (RTL $\rightleftharpoons$ LAND) on the abort path—the design answer to observed silent rejection on ArduCopter 3.3 SITL. Landing interlock: every abnormal termination blocks until the vehicle demonstrably lands and disarms (bounded 240 s) before the queue regains control, and a dequeued mission refuses to arm an armed vehicle—jointly, the queue can never start a flight against an airborne vehicle. Failsafe arbitration: battery, GPS, geofence, and timeout monitors feed a 1 Hz arbiter with monotone severity (LAND never downgraded), N -sample GPS debounce, fire-once event semantics, and mid-RTL escalation. Geofence radii, arrival detection against the 5 m waypoint tolerance, and ETA all use the haversine great-circle distance

$$d = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\varphi}{2} + \cos \varphi_1 \cos \varphi_2 \sin^2 \frac{\Delta\lambda}{2}}, \quad (18)$$

$R = 6371$ km.

The thirteen-state mission FSM within which these mechanisms run is shown in Fig. 3.

v2 adds the response payload. The mission FSM (now thirteen states) records camera evidence during the

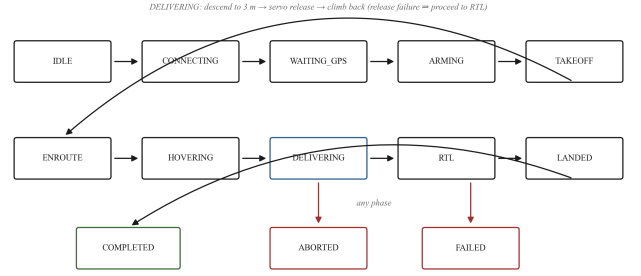


Fig. 3. The 13-state mission FSM, from IDLE through TAKE-OFF, ENROUTE, HOVERING, and DELIVERING to RTL and LANDED, with the abnormal paths into ABORTED and FAILED.

hover window (Pi Camera Module 3 on hardware; a no-op stub in SITL, so the flow is byte-identical) and then enters DELIVERING: descend over the incident point to the configured drop altitude (3.0 m, tolerance 0.7 m, bounded at 45 s), release the first-aid kit via an SG90 servo on Pixhawk AUX OUT 1 commanded with MAV_CMD_DO_SET_SERVO (open 1900 PWM, 2 s settle, re-close 1100), then climb back to cruise altitude for a clean RTL. The 3 m rule is ballistic: a kit released from height h free-falls for

$$t = \sqrt{\frac{2h}{g}} \approx 0.78 \text{ s} \quad \text{at } h = 3 \text{ m}, \quad (19)$$

during which a $v_w = 2$ m/s wind drifts it only

$$x \approx v_w t \approx 1.6 \text{ m}, \quad (20)$$

keeping the kit within reach of the victim—a cruise-altitude release would multiply both drift and impact energy. The governing rule: a failed release is never a reason to loiter—the failure is logged and the drone proceeds to RTL regardless; and the DELIVERING loop polls the failsafe arbiter like every other phase, so a battery or GPS demand pre-empt the drop. All prototype flying is VLOS-only on a registered airframe with an RC safety pilot, per India’s Drone Rules 2021 [5]; autonomous BVLOS response is described strictly as a supervised pilot-program pathway.

VII. Evaluation

A. Methodology

Validation is staged. Tier 1: 68 automated unit cases—47 over the flight stack’s safety logic (every arbiter rule, queue semantics, persistence, every edge-validation bound), 14 over the hub chain (packet seal/unseal, MAC tamper rejection, replay rejection, registry, fusion, pipeline gating—no dispatch below threshold is asserted, not assumed—and dispatcher payload shape), 7 over obstacle keep-out routing. The suites discriminate: run against the pre-hardening implementation, the debounce, cancel, and interlock cases fail. Tier 2: an eight-property SITL

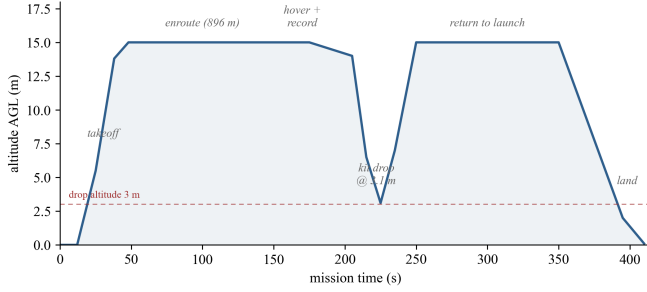


Fig. 4. Altitude profile of the Phase-0 full-chain rehearsal: takeoff to cruise, transit, hover-observe, descent to the 3.1 m kit release, climb-out, and return to launch, completing in 328 s.

acceptance flight (ArduCopter 3.3; 896 m mission at 15 m altitude): simulator up, API up, connected, armed, took off, reached target (≤ 5 m), returned home (≤ 10 m, required), landed. Tier 3: the Phase-0 full-chain rehearsal (scripts/demo_phase0.py)—a synthesized distress WAV stands in for the microphone, a simulated node 600 m from home seals a real packet with the production cryptography, the production hub pipeline verifies (fallback backend), fuses, gates, and dispatches, and the production drone stack flies the SITL mission with hover-record and kit drop. Everything except the audio source, the Stage-2 backend, and the physics is production code.

B. Results

Tier 1: 68/68 pass. Tier 2: 8/8 checks pass; across six acceptance missions spanning the development arc, closest approach was 0.4–0.6 m against the 5 m tolerance and final distance from home 0.0–0.2 m—terminal accuracy bounded by the autopilot’s loiter behaviour, not the dispatch layer. Tier 3: the rehearsal passed end to end—the sealed alert (event scream, PIR active, dark LDR) authenticated and replay-checked; the fallback backend scored the clip above the 0.50 verification threshold; fused severity 0.88 (priority high) cleared the 0.60 dispatch gate; the SITL mission ran the full lifecycle with the recording window active and kit release commanded at 3.1 m relative altitude, completing in 328 s with 0.4 m closest approach. Fig. 4 shows the rehearsal’s altitude profile, with the descent to the 3.1 m release point visible between the hover window and the return leg.

C. What Is Measured vs. In Progress

The Phase-0 result proves architecture and integration: every interface a hardware phase will use—packet bytes, clip convention, thresholds, trigger payload, servo command—was exercised in production form. It deliberately proves nothing about acoustic detection performance. No Stage-1 accuracy exists (the TFLM model is a hook; training and flashing it is Phase 1; <50 ms is a design budget, not a measurement). No field Stage-2 accuracy exists (the Phase-0 score came from the labelled

heuristic fallback on synthesized audio; PANNs’ published performance [6] motivates the backend, not a field claim). For transparency, the fallback scorer is fully specified: over the clip samples it computes the RMS level

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_n x^2[n]}, \quad (21)$$

the spectral centroid

$$C = \frac{\sum_f f |X(f)|}{\sum_f |X(f)|}, \quad (22)$$

and an envelope-burstiness term, combined as

$$\begin{aligned} \text{score} = & 0.45 \min\left(1, \frac{\text{RMS}}{0.15}\right) \\ & + 0.35 \text{clip}\left(\frac{C - 400}{1600}, 0, 1\right) + 0.20 \text{burstiness}. \end{aligned} \quad (23)$$

Equations (21)–(23) describe a labelled development stand-in—loud, high-centroid, bursty audio scores high—not a detection claim. No radio range/loss or ESP-NOW reliability figures exist (Phase 2). No hardware flight has occurred (Phase 3’s staged VLOS progression governs the transition; the SITL firmware is the 2015-vintage 3.3 build, whose command-delivery faults usefully forced the defensive design). These measurements—outdoor detection distance vs. SNR, per-stage latency, end-to-end false-positive rate on street noise, LoRa range vs. spreading factor—are the explicit deliverables of the Phase 1–2 bench campaigns and will appear in the journal version once measured.

VIII. Regulatory, Spectrum, and Privacy Grounding

Deployment legality is designed in, not deferred. **Airspace:** UAS operation in India is governed by The Drone Rules, 2021 (G.S.R. 589(E), Ministry of Civil Aviation) [5], which require registration on the DGCA’s Digital Sky platform and partition airspace into green, yellow, and red zones. All prototype flying sits in the least-restrictive cell of that framework: a registered airframe, VLOS-only, in green-zone private airspace, with an RC safety pilot holding override authority; autonomous BVLOS response is framed strictly as a supervised pilot-program pathway contingent on regulatory approval, which the Rules and DGCA BVLOS experiment schemes provide. **Spectrum:** the prototype’s SX1278 links operate as low-power 433 MHz short-range devices; production would migrate to the WPC-delicensed 865–867 MHz band [11], where higher power is permitted licence-free and Indian LoRa deployments conventionally operate. Neither band needs an operating licence at the power levels used. **Privacy:** the Digital Personal Data Protection Act, 2023 [12] demands purpose limitation, minimisation, and safeguards; the design answers structurally—on-device processing, no continuous recording or transmission, only event-triggered

clips ≤ 5 s retained as incident evidence, and encryption in transit—so compliance is process on top of a minimising architecture, not remediation of an over-collecting one.

IX. Conclusion

VanniKawachh integrates TinyML pole-side screening, pretrained deep audio verification with environmental fusion, sealed operator-free alerting, and a safety-interlocked autonomous first response into a single women-safety chain that asks nothing of the victim but her voice—and demonstrates the complete chain in simulation before any hardware is committed. One discipline governs every layer: a claim is not a confirmation. A Stage-1 hit is not an incident until Stage 2 and fusion say so; a packet is not an alert until its MAC and counter say so; a mode command is not a mode until the autopilot’s telemetry says so; a mission is not finished until the vehicle is disarmed on the ground. Under that discipline the integrated rehearsal passed on the first architecture, and the remaining work is measurement and hardening rather than redesign. Future work: the Phase 1–2 measurement campaigns; live RTSP/WebRTC streaming to police; OpenCV victim tracking during hover; TDOA multi-node localization between poles; and the city-scale node mesh with fleet dispatch.

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Reproducibility and Originality Statement

The full chain is reproducible on any machine with no hardware and no paid service: git clone <https://github.com/SV-1411/drone.git>, a Python 3.10+ environment, pip install -r requirements-dev.txt plus requirements-hub.txt, then python -m pytest (tier 1), python tests/test_full_mission.py (tier 2, boots SITL locally), and python scripts/demo_phase0.py (tier 3, the full-chain rehearsal). All prose was written for this work; protocol, parameter, and flight-mode names are protocol- or firmware-defined identifiers. A large-language-model assistant was used during drafting for structure and consistency; all technical claims derive from the implementation and test logs in the repository, and the final text was reviewed and accepted by the authors.

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